THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a counterexample to the statement "$n^2 + n + 41$ is prime for every natural number n"? 1

(a) $n = 40$ **(b)** $n = 10$ **(c)** $n = 5$ **(d)** $n = 20$

2. The number of factors of 48 is: 1

(a) 9 **(b)** 12 **(c)** 11 **(d)** 10

3. A triangle has sides 10 cm, 24 cm and 26 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) isosceles **(b)** not right-angled **(c)** equilateral **(d)** right-angled

4. For the expression $n^2 + n + 17$, the smallest natural number n for which the value is NOT prime is: 1

(a) 17 **(b)** 18 **(c)** 16 **(d)** 15

5. Many examples that satisfy a statement: 1

(a) prove it for all cases **(b)** show it is false

(c) do not prove it for all cases **(d)** prove its converse

6. Which of the following numbers has an odd number of factors? 1

(a) 54 **(b)** 57 **(c)** 64 **(d)** 50

7. The converse of the proposition "If a quadrilateral is a rhombus, then its diagonals are perpendicular" is: 1

(a) If the diagonals of a quadrilateral are perpendicular, then it is a rhombus.

(b) If it is not true that a quadrilateral is a rhombus, then it is not true that its diagonals are perpendicular.

(c) If a quadrilateral is a rhombus, then its diagonals are perpendicular.

(d) If it is not true that its diagonals are perpendicular, then a quadrilateral is a rhombus.

8. The proposition "If a quadrilateral is a rhombus, then its diagonals are perpendicular" is true. Its converse "If its diagonals are perpendicular, then a quadrilateral is a rhombus" is: 1

(a) True only for some values **(b)** True

(c) It cannot be decided **(d)** False

9. Which of the following is a counterexample to the statement "If $x^2 > 9$, then $x > 3$"? 1

(a) $x = 5$ **(b)** $x = 10$ **(c)** $x = 6$ **(d)** $x = -4$

10. The number of factors of 49 is: 1

(a) 5 **(b)** 2 **(c)** 4 **(d)** 3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Give a counterexample to the statement "$n^2 + n + 41$ is prime for every natural number n". 2

12. How many factors does 72 have? What does the answer tell you about 72? 2

13. Write the converse of "If a triangle is equilateral, then it is isosceles". Is it true? 2

14. Give a counterexample to the statement "The product of two irrational numbers is irrational". 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of "If a triangle is equilateral, then it is isosceles". Is the converse true? Justify. 3

16. Show that the statement "Every odd number is prime" is false by giving a counterexample. 3

17. Use the converse of the Baudhāyana–Pythagoras theorem to decide whether 13 cm, 84 cm, 85 cm are the sides of a right triangle. 3

OR

Write the converse of "If $x = 3$, then $x^2 = 9$". Is the converse true? Justify. 3

18. Show that the statement "If n is prime, then $2n + 1$ is prime" is false by giving a counterexample. 3

OR

Use the converse of the Baudhāyana–Pythagoras theorem to decide whether 14 cm, 48 cm, 50 cm are the sides of a right triangle. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the sum of any 7 consecutive integers is divisible by 7. Verify for the numbers starting at 4. 5

OR

Prove that the square of an odd integer is odd and write the converse. Is the converse true? Test it with 15. 5

20. Prove that the sum of any 3 consecutive integers is divisible by 3. Verify for the numbers starting at 25. 5

OR

Prove that the square of an odd integer is odd and write the converse. Is the converse true? Test it with 9. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND